WebMD Health Services ONE Assessment Review

Background

Health assessments (HAs) have been a key element of best practice workplace health and well-being programs for several decades.[1] They represent unique opportunities to gather information about a population (e.g., employees) that is otherwise unavailable to organizations. With few exceptions, data collected via HA is not available from any other source. Questions typically cover a range of health behaviors, and WebMD's ONE Assessment experience is focused holistically on physical health and overall well-being, including concepts associated with social health, financial health, and emotional health. Participants are asked to rate their level of health as well as their satisfaction with their health. More specifically, we utilize a combination of questions to assess productivity, quality of life/health status, as well as questions on well-being and social connectedness.

Assessing employee health habits is at the core of workplace health and well-being programs and is almost universally viewed as a best practice standard for program implementation because of the valuable information they provide to the person and to the organization.[2,3] The HA is an invaluable tool for tracking and monitoring both individual level and population level health risk prevalence, identifying opportunities for intervention and triaging individuals into appropriate and relevant programs. The best way to maximize the value of the HA is to have it based on current scientific evidence and expert recommendations, using the newest technology.

Development

ONE Assessment has been developed in accordance with evidence-based nationally recognized guidelines and best practices and in collaboration with highly respected researchers and clinical experts, including our Clinical Advisory Board (CAB) and Johns Hopkins University (see below for details). Ron Z. Goetzel, PhD, a member of our CAB and a noted authority and thought leader in HA research and instrument development, has been a long-standing consultant around HA strategy and methodology. Our efforts were informed by widely published results of scientific studies, resulting in a data-driven assessment emphasizing risk measurement, education, and prevention. WebMD references evidence-based research, peer reviewed articles, clinical guidelines, and scientific information from the most appropriate authoritative medical expert groups when reviewing and updating content.

A defined medical editorial review process, both periodic and incidental, is used to review content. At least twice per year, the CAB reviews and updates the group of evidence-based clinical guidelines which are approved for reference for WebMD content. Additionally, all digital content is reviewed in its entirety at least every two years, with appropriate updating of WebMD content and references.

---

[1] U.S. Department of Health and Human Services, Public Health Service. 1992 national survey of worksite health promotion activities: summary. *Am J Health Promot*. 1993;7:452–464.
[2] Terry P, Seaverson EL, Grossmeier J, Anderson D. Association between nine quality components and superior worksite health management program results. J Occup Environ Med. 2008 Jun;50(6):633-641.
[3] Anderson DR, Seaverson ELD, and Terry PE. Chapter 13: Health assessment. In the book: Health Promotion in the Workplace, 2014 (4th) edition. Michael P O'Donnell, Editor. Publisher: Am J Health Promot, Troy, MI. October 1, 2014.

The following is a representative list of professional medical authorities used in developing the science behind our HA:

- American Cancer Society
- American College of Physicians
- American College of Sports Medicine
- American Diabetes Association
- American Heart Association
- American Medical Association
- American Psychiatric Association
- Centers for Disease Control and Prevention
- National Heart, Lung, and Blood Institute
- National Institutes of Health
- U.S. Preventive Services Task Force

WebMD strives to create an inclusive assessment experience that is both meaningful to participants and helpful in addressing their unique clinical risks. We consider the literacy and reading levels, length of the assessment/time to complete and cognitive load of each individual question. Additionally, participants are asked to provide their gender and are presented options including “Male/Man”, “Female/Woman”, “Non-binary”, “Decline to answer” and/or “My gender identity is different than my sex assigned at birth”. The HA leverages participant responses for the display of subsequent questions to ensure a personalized and inclusive experience.

Validation

WebMD commissioned the Institute for Health and Productivity Studies (IHPS) at the Johns Hopkins Bloomberg School of Public Health (JHU) in 2019 to conduct face and content validity of the ONE Assessment. The instrument was further evaluated in terms of its reliability and predictive validity.

Data collected from these validity and reliability assessments were used to further refine the ONE Assessment and create scoring algorithms so that both the content was aligned with current scientific evidence and risk stratification algorithms were consistent with best practices and the best available science.

Accreditation Standards

WebMD's HA is scientifically based, valid and reliable. It has been evaluated and accredited by the National Committee for Quality Assurance (NCQA).

Health Assessment Scoring

Using biometric data, lifestyle information, personal medical history, and readiness-to-change indicators collected with the HA, the system computes risk level/severity for a variety of modifiable risk factors and chronic disease risks, as well as an overall health status score (HA Score).

Health Assessment questions have been derived from validated psychometrics focused on key topics with user responses entered into our scoring algorithms. As always, proprietary, evidence-based health risk scoring algorithms are utilized and are based upon research and field application of national consensus health guidelines and recommendations from subject matter experts in HA methods, health education, medicine, health science and behavior science.

Our scoring methodology has evolved to simplify interpretability and accommodate any future enhancements or modifications. The HA score is derived based on answers related to modifiable health risks (e.g., physical activity, eating practices, etc.) that feed into our proprietary scoring algorithms. Our scoring algorithm targets only modifiable risks because we believe this approach empowers individuals to better manage and control their conditions and improve their overall health and well-being.

The HA Score is on a 100-point scale and is reflective of an individual's overall health status and predictive of a user's perceived health status, future health care utilization, and overall well-being. Indices are created for each domain and then weighted and combined to assess overall risk status. More specifically, individuals are categorized into low, moderate or high risk for each domain. The modifiable risk indices, standardized scores, and their associated weightings are then used to calculate a user's HA Score.

Modifiable Health Risk Factors

Health risk severity levels are calculated for each of the following modifiable risks based on a user's HA responses. WebMD presents modifiable risks in five modules to all users.

- Healthy Behaviors
  - Alcohol use
  - Nutrition
  - Physical Activity
  - Safety
  - Sleep and Fatigue
  - Tobacco Use
- Emotional Health
  - Anxiety
  - Depression
  - Life Satisfaction
  - Social Connectedness
  - Stress
- Work
  - Health and Productivity
- Biometrics
  - Blood Glucose
  - Blood Pressure
  - BMI/Weight
  - Cholesterol
- Clinical
  - Dental
  - Healthcare Utilization
  - Hearing
  - Medication Adherence
  - Pain
  - Prevention and Screenings
  - Vision

Additional Question Sets

ONE Assessment includes questions that allow us to assess health from a holistic well-being perspective. Our goal is to leverage a number of data sources that enable us to better understand each participant's well-being. As such, the ONE Assessment experience was designed to accommodate additional question

sets, as well as client-defined custom questions. The following question sets are available for use within the HA:

- Senior Assessment
- Metabolic Syndrome
- Work Limitations Questionnaire© (short form)
- Social determinants of health
- Financial well-being
- Safety (supplements two core safety questions)

Information gathered from these additional questions can be used to support a more personalized user experience on the WebMD ONE Portal through the Promotional Card tool and for aggregate reporting.

ONE Assessment Reporting

Upon completion of the ONE Assessment, users receive a results report which includes personalized feedback based on health risk status, as well as personalized recommendations and referrals to WebMD health and well-being content, educational resources, and program opportunities, or to client specific programs or resources.

The individual HA results report includes their overall health score as well as risk level (high, moderate, or low) for each health behavior based on the latest scientific guidelines. Health risks are prioritized, which helps participants focus on areas most appropriate for them to change.

ONE Assessment Review

In 2021, WebMD formally established the WebMD HA Taskforce, led by the Center for Research and comprised of a cross-functional internal stakeholders, including Clinical Services, Product Management and UX, and Client Success. The Taskforce was created to ensure that the health assessment:

- **Aligns with diverse client needs**: Ensures our assessments are relevant and effective across various platforms and client requirements.
- **Adapts to evolving global well-being trends**: Keeps our assessment current with the latest well-being research, societal shifts, and global changes.
- **Ensures consistent validation and testing**: Regularly reviews and updates our assessment to maintain its reliability and accuracy.
- **Meets all standards and requirements** as defined by accreditation agencies, including NCQA, URAC and CMS, and to oversee ongoing quality control and accreditation efforts.

- **Meets participants (employees) in their current well-being journey:** Addresses employees' unique well-being needs and stages, guided by our behavior change framework.

The HA Taskforce meets on a quarterly basis to ensure the tool is continuously monitored and evaluated and explore opportunities to evolve the assessment experience. We also partner with our Clinical Advisory Board for expert consulting as appropriate. The Taskforce has established cadenced formal reviews (odd-year cycle). As part of the formal review, the HA Taskforce reviews multiple relevant nationally accepted documents against the WebMD ONE Assessment to ensure the question sets were up to date with the latest published research. As we update our evidence-based guidelines used for the health content of our program, the ONE Assessment is reviewed and updated more frequently as needed. In addition to recently published peer-reviewed studies, several national resources are leveraged for formal reviews, including, but not limited to:

- A Framework for Patient-Centered Health Risk Assessments| Policy, Performance, and Evaluation | CDC
- Interim Guidance for Health Risk Assessments and their Modes of Provision for Medicare Beneficiaries (cms.gov)
- 2023 Program Requirements | CMS
- Assessment Tools | Workplace Health Resources | Tools and Resources | Workplace Health Promotion | CDC
- WebMD Evidence-Based Clinical Guidelines - WHS Health Services - Confluence

WebMD Health Services ONE Assessment (ONE HA) Compared to the Limeade Well-Being Assessment (WBA)

November 2024

Background

Assessing employee health habits is at the core of workplace health and well-being programs and is almost universally viewed as a best practice standard for program implementation because of the valuable information they provide to the person and to the organization. [1,2,3] The WebMD ONE Assessment provides a unique opportunity to gather information about a population (e.g., employees) that is otherwise unavailable to organizations. It is an invaluable tool for tracking and monitoring both holistic well-being at both individual level and population level, identifying opportunities for intervention and triaging individuals into appropriate and relevant programs.

ONE Assessment was developed in accordance with evidence-based nationally recognized guidelines and best practices and in collaboration with highly respected researchers and clinical experts, including our Clinical Advisory Board (CAB) and Johns Hopkins University. Ron Z. Goetzel, PhD, a member of our CAB and a noted authority and thought leader in HA research and instrument development, has been a long-standing consultant around HA strategy and methodology. Our efforts were informed by widely published results of scientific studies, resulting in a data-driven assessment emphasizing risk measurement, education, and prevention. WebMD references evidence-based research, peer reviewed articles, clinical guidelines, and scientific information from the most appropriate authoritative medical expert groups when reviewing and updating content. Additionally, WebMD commissioned the Institute for Health and Productivity Studies (IHPS) at the Johns Hopkins Bloomberg School of Public Health (JHU) in 2019 to conduct face and content validity of the ONE Assessment. The instrument was further evaluated in terms of its reliability and predictive validity. WebMD's HA is scientifically based, valid and reliable. It has been evaluated and accredited by the National Committee for Quality Assurance (NCQA).

Information gathered from these additional questions can be used to support a more personalized user experience on the WebMD ONE Portal which includes personalized recommendations and referrals to WebMD health and well-being content, educational resources, and program opportunities, or to client specific programs or resources. Data gathered from the HA is provided back to clients in aggregated reporting through CoreInsights, cadenced data reviewed, including the year-end annual review to support program design and strategy, identify areas of opportunity and to highlight program impact as it relates to population health and well-being.

---

[1] U.S. Department of Health and Human Services, Public Health Service. 1992 national survey of worksite health promotion activities: summary. *Am J Health Promot*. 1993;7:452–464.
[2] Terry P, Seaverson EL, Grossmeier J, Anderson D. Association between nine quality components and superior worksite health management program results. J Occup Environ Med. 2008 Jun;50(6):633-641.
[3] Anderson DR, Seaverson ELD, and Terry PE. Chapter 13: Health assessment. In the book: Health Promotion in the Workplace, 2014 (4th) edition. Michael P O'Donnell, Editor. Publisher: Am J Health Promot, Troy, MI. October 1, 2014.

Our Health Assessment Taskforce

The WebMD Health Services Center for Research leads the WebMD Health Assessment Taskforce. Our Taskforce is a cross-functional internal stakeholder group, including members across teams and areas of expertise, e.g., multiple members of the Center (researchers, psychometric experts), clinical services, product management, UX, and client services. The Taskforce was created to ensure that the HA:

- Aligns with diverse client needs: Ensures our assessments are relevant and effective across various platforms and client requirements.
- Adapts to evolving global well-being trends: Keeps our assessment current with the latest well-being research, societal shifts, and global changes.
- Ensures consistent validation and testing: Regularly reviews and updates our assessment to maintain its reliability and accuracy.
- Meets all standards and requirements as defined by accreditation agencies, including NCQA, URAC and CMS, and to oversee ongoing quality control and accreditation efforts.
- Meets participants (employees) in their current well-being journey: Addresses employees' unique well-being needs and stages, guided by our behavior change framework.

Our Taskforce meets regularly to discuss opportunities to evolve the current version of the HA. We always welcome client feedback on the HA to ensure that our assessment meets the needs of the diverse clients and employees that we serve.

Table 1. WebMD HA and Limeade WBA Topic-Level Comparison

Concept	WebMD HA	Limeade WBA
Demographics	X	X
Overall Health	X	X
Health Behaviors		
Tobacco Use	X	X
Physical Activity	X	X
Nutrition	X	X
Alcohol Use	X	X
Sleep and Fatigue	X	X
Safety	X	X (health risk)
Emotional Health		
Depression	X	X
Anxiety	X	X
Stress	X	X
Social Connectedness	X	X
Life Satisfaction	X	X
Clinical		
Limitations	X	X
Prevention and Screenings	X	X
Conditions	X	X
Pain	X	X (back pain)
Medication Adherence	X	
Health Care Utilization	X	
Biometrics (e.g., height, weight, cholesterol, etc.)	X	X
Productivity	X	X
Absence	X	
Senior (65+)		
Person filling out the form	X	
Living Situation	X	
Nursing Home/Rehabilitation Visits	X	
Activities of Daily Living (e.g., Transportation, Independence)	X	
Memory	X	
Fall Risk	X	
Sexual Health	X	
Social Services	X	
Financial Well-being	X (optional)	X
Social Determinants of Health	X (optional)	
Safety	X (optional)	X

Table 2. Impact of the Transition from Limeade WBA to WebMD ONE HA

Concept	Advantages of WebMD HA	Measurement Changes
Demographics	Gain more demographic information on gender, language, race, education, marital status, and employment status	
Overall Health		Minor differences in the wording of question
Health Behaviors		
Tobacco Use	Gain more information about tobacco use and desire to quit	
Physical Activity		Minor differences in the questions asked
Nutrition	Gain more detailed information about weekly diet	
Alcohol Use	Gain more information about alcohol consumption and desire to decrease consumption	
Sleep and Fatigue		Minor differences in the questions asked
Safety	Gain information about seat belt and sunscreen use	
Emotional Health		
Depression	Based on PHQ-2	Fewer questions required for an assessment of depression
Anxiety	Based on GAD-2	Fewer questions required for an assessment of anxiety
Stress	Gain information about desire for help managing stress	Minor differences in the questions asked
Social Connectedness		Fewer questions required for an assessment of social connectedness
Life Satisfaction		Fewer questions required for an assessment of life satisfaction
Clinical		
Limitations	Gain information about dental, vision, and hearing	
Prevention and Screenings	Gain more detailed information about specific preventative health screenings	
Conditions	Gain information about medical conditions and treatment	
Pain	Gain information about chronic pain	
Medication Adherence	Gain information about medication adherence	

Health Care Utilization	Gain information about healthcare utilization for the past year	
Biometrics (e.g., height, weight, cholesterol, etc.)	Gain more information about desire for help	
Productivity	Gain information about health-related issues impacting work performance	
Absence	Gain information about number of days absent from work due to health-related issues	
Senior (65+)	Questions specifically geared to assess for the needs of those aged 65 years and older	
Person filling out the form	Gain information about who is filling out the assessment	
Living Situation	Gain information about household	
Weight Loss	Gain information about weight loss	
Prevention and Screenings	Gain information about pneumonia and shingles vaccinations	
Healthcare Utilization	Gain information about stays in a nursing home or rehabilitation center and injuries, surgeries, diagnoses, and medications	
Activities of Daily Living (e.g., Transportation, Independence)	Gain information about transportation needs, daily living activities, and social support	
Memory	Gain information about memory	
Fall Risk	Gain information about fall risk	
Sexual Health (optional)	Gain information about sexual health	
Social Services (optional)	Gain information about interest in receiving social service support	
Financial Well-being (optional)		Minor differences in the questions asked
Social Determinants of Health (optional)	Gain information about social determinants of health, e.g., ability to pay bills, provide food, paying for transportation	
Safety (optional)	Gain information about use of personal safety equipment, repetitive motions, and back protection	Minor differences in the wording of a question about recreational drug use